

Trader Behavior vs Market Sentiment – Project Summary

1. Problem Understanding

The objective was to analyze how trader performance changes under different market sentiments (Fear vs Greed) using historical trade data and a market sentiment index.

2. Data Loading

Two datasets were loaded: one containing individual trade records and another containing daily market sentiment values.

3. Data Cleaning

Column names were standardized, missing values inspected, and data types corrected for consistency. Special attention was given to timestamp columns stored in Unix epoch format.

4. Timestamp Fixing

Trade timestamps were stored in milliseconds, while sentiment timestamps were stored in seconds. Both were correctly converted into datetime format to avoid incorrect 1970 dates.

5. Feature Engineering

Daily dates were extracted from timestamps. Sentiment categories were converted into a binary classification: Fear and Greed. Additional features such as profitability flag and absolute trade size were created.

6. Data Merging

Trades were merged with daily sentiment using a left join to preserve all trade records. Some trades remained unclassified due to missing sentiment dates, which is expected.

7. Exploratory Data Analysis

Trade counts, average PnL, median PnL, and PnL distributions were analyzed across Fear and Greed. Boxplots were used with and without outliers to understand typical and extreme behavior.

8. Key Insights

Fear periods had more trades but lower average profitability. Greed periods showed higher average PnL driven by fewer but larger winning trades. Median PnL was zero in both cases, indicating most trades close near breakeven.